

CO4 - AT2 - Coding Golf Assessment

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QUESTIONS

- Q1**
Bubbled Sorting
25 marks
- Q2**
Insertion Sorting
25 marks
- Q3**
Quick Sorting
25 marks
- Q4**
Merger Sort
25 marks

4/4 answered

Q1 Bubbled Sorting

25 marks

Write a C code for performing Bubbled Sort - n=5 A[] = {8 2 3 5 7}.

Your Code

```

1 #include <stdio.h>
2
3 int main() {
4     int n = 5, a[] = {8, 2, 3, 5, 7}, i, j, t;
5
6     for(i = 0; i < n-1; i++)
7         for(j = 0; j < n-i-1; j++)
8             if(a[j] > a[j+1]) {
9                 t = a[j];
10                a[j] = a[j+1];
11                a[j+1] = t;
12            }
13
14     for(i = 0; i < n; i++)
15         printf("%d ", a[i]);
16
17     return 0;
18 }
```

✔ Submitted

Input

stdin input

Output

2 3 5 7 8

QUESTIONS
Q1 ✓
Bubble Sort
25 marks

Q2 ✓
Insertion Sorting
25 marks

Q3 ✓
Quick Sort
25 marks

Q4 ✓
Merge Sort
25 marks

4/4 answered

Q2 Insertion Sorting

25 marks

Write a C code for performing Insertion Sort - n=5 A[] = {8 2 3 5 7}.

Your Code

```
1 #include <stdio.h>
2
3 int main() {
4     int a[] = {8,2,3,5,7}, n=5, i, j, key;
5
6     for(i=1;i<n;i++) {
7         key=a[i];
8         j=i-1;
9         while(j>=0 && a[j]>key) {
10             a[j+1]=a[j];
11             j--;
12         }
13         a[j+1]=key;
14     }
15
16     for(i=0;i<n;i++)
17         printf("%d ",a[i]);
18
19     return 0;
20 }
```

Submitted

Input

stdin input

Output

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Quick Sorting
25 marks

Q4
Merger Sort
25 marks

4/4 answered

Q3 Quick Sorting

25 marks

Write a C code for performing Quick Sort - $n=7$ $A[] = \{8, 2, 3, 5, 7, 1, 6\}$.

Your Code

```
1 #include <stdio.h>
2
3 void quicksort(int a[], int low, int high) {
4     int i = low, j = high, p = a[(low + high) / 2], t;
5
6     while (i <= j) {
7         while (a[i] < p) i++;
8         while (a[j] > p) j--;
9
10        if (i <= j) {
11            t = a[i];
12            a[i] = a[j];
13            a[j] = t;
14            i++;
15            j--;
16        }
17    }
18
19    if (low < j) quicksort(a, low, j);
20    if (i < high) quicksort(a, i, high);
21 }
```

Input
 stdin input

Output

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QUESTIONS
Q1
Bubbled Sorting
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25 marks

Q3
Quick Sorting
25 marks

Q4
Merger Sort
25 marks

4/4 answered

Q4 Merger Sort

25 marks

Write a C code for performing Merger Sort - n=8 A[] = {8 2 3 5 7 6 4 1}.

Your Code

```
1 #include <stdio.h>
2
3 void merge(int a[], int l, int m, int r) {
4     int i=l, j=m+1, k=0, t[20];
5
6     while(i<=m && j<=r) {
7         if(a[i]<a[j]) t[k++]=a[i++];
8         else t[k++]=a[j++];
9     }
10
11     while(i<=m) t[k++]=a[i++];
12     while(j<=r) t[k++]=a[j++];
13
14     for(i=0; i<k; i++)
15         a[l+i]=t[i];
16 }
17
18 void mergeSort(int a[], int l, int r) {
19     if(l<r) {
20         int m=(l+r)/2;
21         mergeSort(a,l,m);
22         mergeSort(a,m+1,r);
23         merge(a,l,m,r);
24     }
25 }
```

Input

stdin input

Output